

Minimax Estimation

A Crash Course — plus a Compendium of Every Theorem

Parametric Inference, B.Stat. 3rd Year — Lectures 8–9, and all ten lectures

Third and last of the crash courses, after `PI_Bayes_Crash_Course.pdf` and `PI_Testing_Crash_Course.pdf`.

Part I builds minimax from scratch; **Part II** is a reference containing every theorem in the course, each with its hypotheses called out separately — because in this subject the marks are in the hypotheses, not the statement.

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Part I — Minimax from scratch

1 The third escape route

The same starting point as always. The risk $R(\theta, T) = \mathbb{E}_\theta(T - \theta)^2$ is a *function* of θ , so “minimise the risk” is not a well-posed problem. Three ways to make it well posed, and you now know all three:

	How the function becomes a number	Character
UMVUE	Don't collapse it — shrink the class of T instead	Optimistic: assumes unbiasedness is a reasonable restriction
Bayes	$r(\pi, T) = \int R(\theta, T)\pi(\theta)d\theta$ — <i>average</i> it	Requires you to commit to weights π
Minimax	$\sup_\theta R(\theta, T)$ — take the <i>worst case</i>	Pessimistic: protects against the least favourable θ , commits to nothing

Definition

T^* is **minimax** if

$$\sup_{\theta \in \Theta} R(\theta, T^*) = \inf_T \sup_{\theta \in \Theta} R(\theta, T).$$

Read it right to left: among all estimators, look at each one's worst-case risk, and pick the estimator whose worst case is least bad. The common value is the **minimax value** of the problem.

2 Why minimax is hard, and the one trick that solves it

Suppose someone hands you a candidate T^* . Computing $\sup_\theta R(\theta, T^*)$ is easy — it is calculus. But proving T^* is *minimax* means showing

$$\sup_\theta R(\theta, T) \geq \sup_\theta R(\theta, T^*) \quad \text{for every estimator } T,$$

and there are infinitely many T , most of which you cannot write down. You cannot check them one at a time. You need a **certificate**: a lower bound on $\inf_T \sup_\theta R(\theta, T)$ that you can compute, and that happens to equal $\sup_\theta R(\theta, T^*)$.

Where lower bounds come from — this is the whole subject

For **any** probability measure π on Θ and **any** estimator T ,

$$\sup_\theta R(\theta, T) \geq \int_\Theta R(\theta, T) \pi(\theta) d\theta = r(\pi, T),$$

simply because a supremum is at least any weighted average. Now minimise both sides over T :

$$\inf_T \sup_\theta R(\theta, T) \geq \inf_T r(\pi, T) = r(\pi, T_\pi),$$

the Bayes risk of the Bayes estimate for π .

So every prior π hands you a free lower bound on the minimax value, namely its Bayes risk. That is why Bayes estimation is taught immediately before minimax — it

is not a detour, it is the machine that produces the certificates.

The sandwich

Putting it together, for any π and any T :

$$\underbrace{\sup_{\theta} R(\theta, T^*)}_{\text{what you can compute}} \geq \underbrace{\inf_T \sup_{\theta} R(\theta, T)}_{\text{what you want}} \geq \underbrace{r(\pi, T_{\pi})}_{\text{a computable lower bound}}.$$

The left inequality is trivial (T^* is one of the T 's). **If you can find a π making the two outer quantities equal, the sandwich collapses and T^* is minimax.** Everything in Lectures 8–9 is a recipe for finding that π .

A π achieving the best possible lower bound is called a **least favourable prior**: it puts its mass where estimation is hardest, making the problem as difficult as it can be.

3 The two theorems, and when to use which

Theorem A — constant-risk (equalizer) criterion

If T_{π} is the Bayes estimate for a **proper** prior π , and $R(\theta, T_{\pi}) = c$ is **constant in θ** , then T_{π} is minimax.

Proof: Substitute into the sandwich. For any T ,

$$\sup_{\theta} R(\theta, T) \geq r(\pi, T) \geq r(\pi, T_{\pi}) = \int c \pi(\theta) d\theta = c = \sup_{\theta} R(\theta, T_{\pi}). \quad \square$$

The constant risk is what makes step three work: the average of a constant is that constant, so the lower bound lands exactly on T_{π} 's worst case.

Theorem B — limit of Bayes rules

Let $\{\pi_k\}$ be proper priors with Bayes risks $r(\pi_k, T_{\pi_k}) \rightarrow c$. If T^* satisfies $\sup_{\theta} R(\theta, T^*) = c$, then T^* is minimax.

Proof: For any T and every k , $\sup_{\theta} R(\theta, T) \geq r(\pi_k, T) \geq r(\pi_k, T_{\pi_k})$. Let $k \rightarrow \infty$: $\sup_{\theta} R(\theta, T) \geq c = \sup_{\theta} R(\theta, T^*)$. $\square$

Common confusion: “Why do we need Theorem B at all?”

Because **the minimax estimator is often not Bayes for any proper prior**, so Theorem A simply does not apply.

The example is $X \sim N(\theta, 1)$ with $\theta \in \mathbb{R}$. The minimax estimator is X itself, with constant risk 1. But is X Bayes for some proper prior? No: every proper prior pulls the estimate toward its own mean, so every proper Bayes estimate is a strict shrinkage $\frac{k}{k+1}X$, never X . Only the *improper* flat prior gives X , and an improper prior has infinite Bayes risk, so the “sup $\geq$ average” step is meaningless.

The repair is Theorem B: use proper priors $\pi_k = N(0, k)$ that approach flatness, and let their Bayes risks $k/(k+1)$ climb to 1. The sequence $\{\pi_k\}$ is called a *least favourable sequence*.

Rule of thumb: bounded $\Theta \rightarrow$ try Theorem A; unbounded $\Theta \rightarrow$ you will almost certainly need Theorem B.

Why minimax rules tend to have constant risk

Not a theorem, but the right intuition. Suppose $R(\cdot, T)$ is not flat, peaking at some θ_1 . Intuitively you can nudge T to do a little worse where the risk is low and a little better near θ_1 , lowering the peak — and the peak is all that minimaxity cares about. At the optimum there should be nothing left to shave, i.e. the risk is flat.

An estimator with constant risk is called an **equalizer**. So the practical search strategy is: *look for an equalizer inside a conjugate family, then check it is Bayes*. That is exactly how the binomial answer is found below.

4 Worked example 1: the Binomial (Theorem A in full)

$X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$. Here Θ is bounded, so try Theorem A.

Step 1. Get a family of Bayes estimates. With prior $\text{Beta}(p, q)$ the posterior is $\text{Beta}(p + x, q + n - x)$, so writing $c = p + q + n$,

$$T_{p,q} = \frac{p + X}{c}.$$

Step 2. Compute its risk. $\mathbb{E}_\theta X = n\theta$ and $\text{Var}_\theta X = n\theta(1 - \theta)$, so

$$\text{Bias} = \frac{p + n\theta}{c} - \theta = \frac{p - (p + q)\theta}{c}, \quad \text{Var}(T_{p,q}) = \frac{n\theta(1 - \theta)}{c^2},$$

$$R(\theta, T_{p,q}) = \frac{1}{c^2} \left[n\theta(1 - \theta) + (p - (p + q)\theta)^2 \right].$$

Step 3. Force it to be constant. Expand the bracket as a quadratic in θ :

$$\underbrace{[(p + q)^2 - n]}_{\text{coeff of } \theta^2} \theta^2 + \underbrace{[n - 2p(p + q)]}_{\text{coeff of } \theta} \theta + \underbrace{p^2}_{\text{constant}}.$$

Both non-constant coefficients must vanish:

$$(p + q)^2 = n \Rightarrow p + q = \sqrt{n}; \quad n = 2p(p + q) = 2p\sqrt{n} \Rightarrow p = \frac{\sqrt{n}}{2}, \quad q = \frac{\sqrt{n}}{2}.$$

This is the *only* solution — two equations, two unknowns.

Step 4. Read off the answer. With $c = n + \sqrt{n} = \sqrt{n}(1 + \sqrt{n})$,

$$T^* = \frac{X + \sqrt{n}/2}{n + \sqrt{n}}, \quad R(\theta, T^*) = \frac{p^2}{c^2} = \frac{n/4}{n(1 + \sqrt{n})^2} = \frac{1}{4(1 + \sqrt{n})^2}$$

for every θ . It is Bayes for a proper prior and has constant risk, so by Theorem A it is **minimax**, and $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ is least favourable.

Sanity check on the least favourable prior

$\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ has mean $\frac{1}{2}$ and variance $\frac{n/4}{n(\sqrt{n} + 1)} = \frac{1}{4(\sqrt{n} + 1)} \rightarrow 0$. So as n grows the least favourable prior **concentrates at** $\theta = \frac{1}{2}$ — exactly where $\text{Var}_\theta(X) = n\theta(1 - \theta)$ is largest, i.e. where estimation is hardest. The certificate is being placed precisely where the problem is worst, which is what “least favourable” should mean.

4.1 Minimax versus the UMVUE — and what minimaxity actually buys

The UMVUE is X/n , with $R(\theta, X/n) = \theta(1-\theta)/n$ and $\max_{\theta} R = \frac{1}{4n}$ at $\theta = \frac{1}{2}$. Since $(1 + \sqrt{n})^2 = n + 2\sqrt{n} + 1 > n$,

$$\frac{1}{4(1 + \sqrt{n})^2} < \frac{1}{4n} \implies \text{the UMVUE is not minimax.}$$

But look at the whole risk curves, say $n = 100$:

	$\theta = 0.5$	$\theta = 0.1$	$\theta \rightarrow 0$
UMVUE X/n	0.00250	0.00090	$\rightarrow 0$
Minimax T^*	0.002066	0.002066	0.002066

T^* wins at the peak and **loses badly everywhere else** — near the endpoints the UMVUE has almost no risk while T^* still pays its flat premium.

Common confusion: “So is the minimax estimator the best one?”

No. Minimaxity is a **worst-case** guarantee, not a uniform one. T^* is the right choice only if you genuinely fear an adversary choosing the worst θ . If you have any reason to think θ is near 0 or 1, the UMVUE is far better. Note also the ratio $\frac{(1+\sqrt{n})^2}{n} \rightarrow 1$: for large n the two are almost identical anyway, so the whole distinction is a small-sample phenomenon.

5 Worked example 2: the Normal (Theorem B in full)

$X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$ — now Θ is unbounded.

Step 1. Candidate and its worst-case risk. $R(\theta, X) = \mathbb{E}_{\theta}(X - \theta)^2 = 1$ for every θ , so X is an equalizer with $\sup_{\theta} R(\theta, X) = 1$.

Step 2. Build the certificate sequence. Take $\pi_k = N(0, k)$. Completing the square,

$$\theta \mid X = x \sim N\left(\frac{k}{k+1}x, \frac{k}{k+1}\right), \quad T_{\pi_k} = \frac{k}{k+1}X.$$

The posterior variance is the *constant* $\frac{k}{k+1}$, so by $r(\pi, T_{\pi}) = \mathbb{E}_m[\text{Var}(\theta \mid X)]$,

$$r(\pi_k, T_{\pi_k}) = \frac{k}{k+1}.$$

Step 3. Close the sandwich. $r(\pi_k, T_{\pi_k}) \rightarrow 1 = \sup_{\theta} R(\theta, X)$, so by Theorem B, X is **minimax** and the minimax value is 1.

What the sequence is doing

$\pi_k = N(0, k)$ spreads out as k grows, approaching the (improper) flat prior. Each π_k gives a valid lower bound $k/(k+1)$; none of them reaches 1, but together they climb to it. No *single* prior certifies X ; the sequence does. That is the entire reason Theorem B exists, and it is Homework 2 Q1.

6 Worked example 3: when the minimax value is infinite

$X \sim \text{Poisson}(\lambda)$, $\lambda \in (0, \infty)$. Take $\pi = \text{Gamma}(a, \text{scale } b)$. The posterior is $\text{Gamma}(a + x, \frac{b}{1+b})$ with variance $\frac{(a+x)b^2}{(1+b)^2}$, and the marginal mean of X is $\mathbb{E}[\lambda] = ab$, so

$$r(\pi_{a,b}) = \mathbb{E}_m[\text{Var}(\lambda | X)] = \frac{b^2(a + ab)}{(1+b)^2} = \frac{ab^2}{1+b}.$$

Take $a = 1$ and $b = k \rightarrow \infty$: $r = \frac{k^2}{1+k} \rightarrow \infty$. Since every estimator's worst-case risk dominates every Bayes risk,

$$\sup_{\lambda > 0} R(\lambda, T) \geq \frac{k^2}{1+k} \rightarrow \infty \quad \text{for every } T.$$

So **every** estimator has infinite maximum risk and no minimax estimator exists in any useful sense (Homework 2 Q2).

The pattern to recognise

Infinite minimax value happens when Θ is **unbounded** and the achievable variance grows without bound along it. Two instances in the course:

- $\text{Poisson}(\lambda)$, $\lambda \in (0, \infty)$: $\text{Var}_\lambda(X) = \lambda \rightarrow \infty$.
- $U[a, \theta]$, $\theta \geq a$ unbounded: with Pareto priors of scale m , the Bayes risk is $\geq C(\alpha, n)m^2 \rightarrow \infty$.

Contrast $\text{Bin}(n, \theta)$ with $\theta \in (0, 1)$ bounded, and $N(\theta, 1)$ where Θ is unbounded but the variance is the *constant* 1. **Unboundedness alone is not fatal — it has to be paired with unbounded variance.**

7 The recipe

Given a minimax question, do this

1. **Is Θ bounded?** If not, first ask whether the variance blows up along it — if so, show every estimator has infinite maximum risk (Poisson pattern) and stop.
2. **Compute** $\sup_\theta R(\theta, T^*)$ for your candidate. If the risk is constant, you have an equalizer and are in good shape.
3. **Is T^* Bayes for a proper prior?** Search inside the conjugate family: write down $R(\theta, T_{\text{prior params}})$, expand as a polynomial in θ , and force the non-constant coefficients to vanish. If a proper prior works, quote **Theorem A** and finish in one line.
4. **If no proper prior gives T^*** (typical when T^* is generalized Bayes under an improper prior), build a sequence π_k of proper priors with $r(\pi_k) \rightarrow \sup_\theta R(\theta, T^*)$ and quote **Theorem B**.
5. **Say which prior (or sequence) is least favourable.** It is usually worth a mark, and it shows you understand what the certificate is.

Problem	Minimax rule	Minimax value	Certificate
$\text{Bin}(n, \theta)$	$\frac{X + \sqrt{n}/2}{n + \sqrt{n}}$	$\frac{1}{4(1 + \sqrt{n})^2}$	Theorem A with $\text{Beta}(\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2})$
$N(\theta, 1)$	X	1	Theorem B with $\pi_k = N(0, k)$
$N(\theta, \sigma_0^2), n \text{ obs}$	$\bar{X}$	σ_0^2/n	Theorem B, same idea
$\text{Poi}(\lambda)$	none	∞	Gamma priors, $r \rightarrow \infty$
$U[a, \theta]$	none	∞	Pareto priors, $r \rightarrow \infty$

Part II — Every theorem in the course

8 Quick reference

The hypotheses column is what examiners actually check. Quoting a theorem without verifying its hypotheses is the most reliable way to lose marks in this subject.

	Theorem	Hypotheses to verify	Gives you
T1	Neyman–Fisher factorisation	$f = g(\theta, T)h(\mathbf{x})$, h free of θ	T sufficient (and conversely)
T2	Rao–Blackwell	S sufficient; $\mathbb{E}T^2 < \infty$	$\mathbb{E}[T S]$ has same bias, no more variance
T3	Lehmann–Scheffé	S complete sufficient; $h(S)$ unbiased	$h(S)$ is <i>the</i> UMVUE, a.s. unique
T4	UMVUE uniqueness	a UMVUE exists	it is a.s. unique (no completeness needed)
T5	Exponential-family completeness	natural parameter space contains an open interval (rectangle if k -dim)	$\sum_i T(X_i)$ complete sufficient
T6	Basu	S complete sufficient; V ancillary	$S \perp\!\!\!\perp V$
T7	Cramér–Rao	regularity: support free of θ , differentiation under $\int$ legal	$\text{Var}(T) \geq [\psi']^2/(nI)$
T8	CRLB attainment	score $= k(\theta)[T - \psi(\theta)]$	equality, iff exp. family
T9	Bayes estimate	squared error loss	posterior mean; $r = \mathbb{E}_m[\text{Var}(\theta X)]$
T10	Constant-risk minimax	T_π Bayes for a proper π ; risk constant	T_π minimax
T11	Limit of Bayes rules	π_k proper; $r(\pi_k) \rightarrow c = \sup_\theta R(\theta, T^*)$	T^* minimax
T12	Neyman–Pearson	both hypotheses simple	MP test is the LR test
T13	Exp. family $\Rightarrow$ MLR	$c(\theta)$ monotone	MLR in $T(x)$
T14	Karlin–Rubin	MLR in T ; one-sided H_A	$\{T > c\}$ is UMP; power monotone
T15	Consistency via Chebyshev	bias $\rightarrow 0$ and Var $\rightarrow 0$	$T_n \xrightarrow{p} \psi(\theta)$
T16	MLE consistency	identifiability, common support, uniform LLN, unique root	$\hat{\theta}_n \xrightarrow{p} \theta_0$

9 Estimation

T1. Neyman–Fisher factorisation

Hypotheses (state and check these): $X_1, \dots, X_n$ have joint pdf/pmf $f(\mathbf{x} | \theta)$.

Conclusion: T is sufficient for $\theta \iff f(\mathbf{x} | \theta) = g(\theta, T(\mathbf{x})) h(\mathbf{x})$ with h free of θ .

Used for: Every sufficiency claim in the course. Also proves that the posterior, the MLE and every generalized Bayes estimate depend on the data only through T .

T2. Rao–Blackwell

Hypotheses (state and check these): S sufficient; T an estimator with $\mathbb{E}_\theta T^2 < \infty$.

Conclusion: $T^* = \mathbb{E}[T \mid S]$ is a statistic, $\mathbb{E}_\theta T^* = \mathbb{E}_\theta T$, and $R(\theta, T^*) \leq R(\theta, T)$ for all θ , strictly unless T is already a function of S .

Proof: Sufficiency makes $\mathbb{E}[T \mid S]$ free of θ , hence a statistic. Tower property gives equal means, hence equal bias. Then $\text{Var}(T) = \mathbb{E}[\text{Var}(T \mid S)] + \text{Var}(\mathbb{E}[T \mid S]) \geq \text{Var}(T^*)$, the first term being ≥ 0 and zero iff T is S -measurable. Equal bias plus smaller variance gives smaller MSE. $\square$

Used for: Constructing UMVUEs by conditioning a crude unbiased estimator (typically an indicator).

T3. Lehmann–Scheffé

Hypotheses (state and check these): S complete and sufficient; $h(S)$ unbiased for $\psi(\theta)$.

Conclusion: $h(S)$ is the UMVUE of $\psi(\theta)$, a.s. unique.

Proof: For any unbiased T , $T^* = \mathbb{E}[T \mid S]$ is unbiased, a function of S , and has $\text{Var}(T^*) \leq \text{Var}(T)$ by T2. Both T^* and $h(S)$ are unbiased functions of S , so $\mathbb{E}_\theta[T^* - h(S)] = 0$ for all θ and completeness forces $T^* = h(S)$ a.s. Hence $\text{Var}(h(S)) \leq \text{Var}(T)$. $\square$

Used for: The workhorse. **Completeness is the hypothesis people forget** — without it the conclusion is false, as $U[\theta, \theta + 1]$ shows.

T4. Uniqueness of the UMVUE

Hypotheses (state and check these): A UMVUE exists.

Conclusion: It is a.s. unique. (No complete sufficient statistic required — this is independent of T3.)

Proof: Let T_1, T_2 be UMVUEs, both with variance σ^2 . Then $T = \frac{1}{2}(T_1 + T_2)$ is unbiased, so $\sigma^2 \leq \text{Var}(T) = \frac{1}{4}[\sigma^2 + \sigma^2 + 2\rho\sigma^2] = \frac{\sigma^2}{2}(1 + \rho)$, giving $\rho \geq 1$; but $\rho \leq 1$, so $\rho = 1$. Perfect correlation forces $T_2 = aT_1 + b$ with $a > 0$, and matching means and variances gives $a = 1$, $b = 0$. $\square$

T5. Completeness of exponential families

Hypotheses (state and check these): $f(x \mid \theta) = p(\theta)h(x) \exp\{c(\theta)T(x)\}$ with support free of θ , and the natural parameter space $\Gamma = \{c(\theta) : \theta \in \Theta\}$ contains an interval with non-empty interior. (k -parameter version: a k -dimensional rectangle with non-empty interior.)

Conclusion: $S = \sum_i T(X_i)$ is complete sufficient.

Used for: Establishing completeness in one line for Bin, Poi, Exp, Normal, Gamma, power-series and Multinomial. **The hypothesis bites** for $N(\theta, \theta^2)$, where the natural parameters trace a curve, not a rectangle.

T6. Basu

Hypotheses (state and check these): S complete sufficient for θ ; V ancillary (distribution free of θ).

Conclusion: $S \perp\!\!\!\perp V$ under every P_θ .

Proof: Let $g(s) = \mathbb{P}(V \leq v \mid S = s)$, free of θ by sufficiency. Then $\mathbb{E}_\theta[g(S)] = \mathbb{P}_\theta(V \leq v) = \mathbb{P}(V \leq v)$, a constant free of θ by ancillarity. So $\mathbb{E}_\theta[g(S) - \mathbb{P}(V \leq v)] = 0$ for all θ , and completeness gives $g(S) = \mathbb{P}(V \leq v)$ a.s. That is exactly independence. $\square$

Used for: Proving $\bar{X} \perp\!\!\!\perp S^2$; independence of $\sum X_i$ and scale-invariant ratios (HW1 Q6); and the range identity $\mathbb{E}[R] = \mathbb{E}[V]\mathbb{E}[S]$. **Spotting ancillarity:** differences are ancillary in location families, ratios in scale families.

10 Information

T7. Cramér–Rao lower bound

Hypotheses (state and check these): Regularity: the support does not depend on θ , and differentiation under the integral sign is legal.

Conclusion: For any T with $\mathbb{E}_\theta T = \psi(\theta)$, $\text{Var}_\theta(T) \geq \frac{[\psi'(\theta)]^2}{n I(\theta)}$, where $I(\theta) = \text{Var}_\theta \left(\frac{\partial}{\partial \theta} \log f \right) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f \right]$.

Used for: A benchmark, and an easy way to *prove* something is the UMVUE when the bound is attained. **Meaningless for** $U[0, \theta]$ — the support moves, regularity fails, and $\frac{n+1}{n} X_{(n)}$ has variance of order θ^2/n^2 , far below any naive bound.

T8. Attainment of the CRLB

Conclusion: Equality holds at every $\theta \iff$ the score factorises as $\frac{\partial}{\partial \theta} \log f(\mathbf{x} | \theta) = k(\theta)[T(\mathbf{x}) - \psi(\theta)]$, i.e. iff the model is a one-parameter exponential family and ψ is the mean of its natural sufficient statistic.

Used for: Deciding attainment questions instantly (HW1 Q4: yes, for $\mathbb{E}X$ in a power-series family; Mock Q3: no, for $e^{-\lambda}$ in the Poisson).

Common confusion: failing the CRLB does not disqualify a UMVUE

The CRLB bounds *all* unbiased estimators. If no estimator attains it, the best one still sits strictly above. “Its variance exceeds the CRLB, so it is not the UMVUE” is the single most common error on this topic.

11 Bayes and minimax

T9. Bayes estimate under squared error loss

Hypotheses (state and check these): Squared error loss. (Absolute error gives the posterior median; 0–1 loss gives the mode.)

Conclusion: $T_\pi(x) = \mathbb{E}[\theta | X = x]$, the posterior mean, and $r(\pi, T_\pi) = \mathbb{E}_m[\text{Var}(\theta | X)]$.

Proof: Write $f(x | \theta)\pi(\theta) = m(x)\pi(\theta | x)$ and swap the order of integration: $r(\pi, T) = \int m(x) \left[\int (T(x) - \theta)^2 \pi(\theta | x) d\theta \right] dx$. For each fixed x the inner bracket is minimised over the number $T(x)$ at the posterior mean, with minimum value the posterior variance. $\square$

Used for: Every Bayes computation, and — via the risk formula — every minimax certificate.

T10. Constant-risk minimax criterion

Hypotheses (state and check these): T_π Bayes for a **proper** prior π ; $R(\theta, T_\pi) = c$ constant in θ .

Conclusion: T_π is minimax; π is least favourable.

Proof: $\sup_\theta R(\theta, T) \geq r(\pi, T) \geq r(\pi, T_\pi) = c = \sup_\theta R(\theta, T_\pi)$. $\square$

Used for: The Binomial minimax estimator. **Properness is essential** — under an improper prior the Bayes risk is infinite and the middle step collapses.

T11. Limit of Bayes rules

Hypotheses (state and check these): $\{\pi_k\}$ proper priors with $r(\pi_k, T_{\pi_k}) \rightarrow c$; $\sup_\theta R(\theta, T^*) = c$.

Conclusion: T^* is minimax; $\{\pi_k\}$ is a least favourable sequence.

Proof: $\sup_{\theta} R(\theta, T) \geq r(\pi_k, T) \geq r(\pi_k, T_{\pi_k})$ for every k ; let $k \rightarrow \infty$. $\square$

Used for: X minimax for $N(\theta, 1)$ (HW2 Q1); and, run in reverse, proving the minimax value is ∞ for the Poisson (HW2 Q2) and for $U[a, \theta]$.

12 Testing

T12. Neyman–Pearson lemma

Hypotheses (state and check these): Both hypotheses **simple**: $H_0 : X \sim f_0$ vs $H_A : X \sim f_1$.

Conclusion: There exist $k \geq 0, \gamma \in [0, 1]$ with $\phi^* = 1, \gamma, 0$ according as $f_1 \geq k f_0$, having $\mathbb{E}_0 \phi^* = \alpha$; any such ϕ^* is most powerful of its size; and any MP level- α test has this form a.e. off $\{f_1 = k f_0\}$.

Proof: On $\{f_1 > k f_0\}$, $\phi^* = 1 \geq \phi$; on $\{f_1 < k f_0\}$, $\phi^* = 0 \leq \phi$; on $\{f_1 = k f_0\}$ the second factor vanishes. So $(\phi^* - \phi)(f_1 - k f_0) \geq 0$ pointwise. Integrating, $0 \leq (\mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi) - k(\mathbb{E}_0 \phi^* - \mathbb{E}_0 \phi) \leq \mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi$. $\square$

Used for: Every MP test. Remember the region is one-sided *in* Λ , not necessarily in x — see the Laplace-vs-normal annulus (HW2 Q5) and the non-monotone ordering in HW3 Q3.

T13. Exponential family $\Rightarrow$ MLR

Hypotheses (state and check these): $f(x | \theta) = p(\theta)h(x)e^{c(\theta)T(x)}$ with c non-decreasing.

Conclusion: The family has MLR in $T(x)$.

Proof: $\frac{f_{\theta_1}}{f_{\theta_0}} = \frac{p(\theta_1)}{p(\theta_0)} \exp\{[c(\theta_1) - c(\theta_0)]T(x)\}$, non-decreasing in $T(x)$ since the exponent coefficient is ≥ 0 . $\square$

Used for: Establishing MLR for Bin, Poi, Exp, $N(\theta, 1)$, $N(0, \sigma^2)$ (in x^2). $U[0, \theta]$ has MLR in $X_{(n)}$ but must be checked *directly* — it is not an exponential family. The **Cauchy has no MLR**: the ratio of two quadratics tends to 1 at both ends, so it cannot be monotone.

T14. Karlin–Rubin

Hypotheses (state and check these): MLR in T ; hypotheses **one-sided**, $H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$.

Conclusion: $\phi^* = 1, \gamma, 0$ according as $T \geq c$, with $\mathbb{E}_{\theta_0} \phi^* = \alpha$, is UMP of level α ; and $\beta_{\phi^*}(\theta)$ is non-decreasing, so the size is attained at θ_0 .

Proof: sketch: By NP the MP test against any fixed $\theta_1 > \theta_0$ rejects for large $f_{\theta_1}/f_{\theta_0}$, hence for large T by MLR; the cutoff is determined by $\mathbb{P}_{\theta_0}(T > c) = \alpha$, which does not involve θ_1 . So the same test is MP against every $\theta_1 > \theta_0$. Monotone power upgrades the null from $= \theta_0$ to $\leq \theta_0$ at no cost in size. $\square$

Used for: All one-sided UMP tests (HW3 Q1, Q2; Mock Q8). **Randomise if T is discrete.**

Common confusion: two-sided alternatives

No UMP exists in a regular MLR family: the best test against $\theta_1 > \theta_0$ rejects large T , against $\theta_1 < \theta_0$ rejects small T . **Exception:** if the support moves with θ , the impossible-under- H_0 region is free, and a two-sided UMP test can exist — $U[0, \theta]$ with $H_0 : \theta = 1$ (HW3 Q4).

13 Asymptotics

T15. Consistency via Chebyshev

Hypotheses (state and check these): $\mathbb{E}_\theta T_n \rightarrow \psi(\theta)$ and $\text{Var}_\theta(T_n) \rightarrow 0$.

Conclusion: $T_n \xrightarrow{p} \psi(\theta)$.

Proof: $\mathbb{P}(|T_n - \psi| > \varepsilon) \leq \varepsilon^{-2} \mathbb{E}(T_n - \psi)^2 = \varepsilon^{-2} [\text{bias}^2 + \text{Var}] \rightarrow 0$. $\square$

Used for: Almost every consistency question. **Sufficient, not necessary** — $X_{(n)}$ for $U[0, \theta]$ is biased at every n and consistent; and an estimator with no moments at all can be consistent.

T16. Consistency of the MLE (Wald)

Hypotheses (state and check these): Identifiability; support free of θ ; Θ compact (or the maximiser eventually in a compact set); a uniform law of large numbers for $\log f$.

Conclusion: $\hat{\theta}_n \xrightarrow{p} \theta_0$.

Proof: sketch: By Jensen, $M(\theta) - M(\theta_0) = \mathbb{E}_{\theta_0} \log \frac{f(X|\theta)}{f(X|\theta_0)} \leq \log \int f(x|\theta) dx = 0$, with equality only at θ_0 by identifiability — so the population criterion M has a strict maximum at θ_0 . The sample criterion $M_n = \frac{1}{n} \sum \log f(X_i|\theta)$ converges to M uniformly, so its maximiser converges to θ_0 . $\square$

Used for: **Regularity is sufficient, not necessary:** $U[0, \theta]$ violates the common-support hypothesis yet $X_{(n)}$ is consistent (and n -consistent, not merely $\sqrt{n}$ -consistent).

T17. Consistency of UMVUEs

Hypotheses (state and check these): T_n is the UMVUE of $\psi(\theta)$ for each n , and *some* unbiased sequence $\tilde{T}_n$ has $\text{Var}(\tilde{T}_n) \rightarrow 0$.

Conclusion: $T_n \xrightarrow{p} \psi(\theta)$.

Proof: $0 \leq \text{Var}(T_n) \leq \text{Var}(\tilde{T}_n) \rightarrow 0$ by minimality; T_n is unbiased, so Chebyshev applies. $\square$

14 Two results that are not called theorems but behave like them

Estimability in the Binomial. $\psi(\theta)$ has an unbiased estimator based on $X \sim \text{Bin}(m, \theta)$ iff ψ is a polynomial of degree $\leq m$. ($\Rightarrow$: $\mathbb{E}_\theta T$ is such a polynomial, and two polynomials agreeing on an interval are identical. $\Leftarrow$: X^k/m^k is unbiased for θ^k .)

Estimability in the Poisson. ψ is estimable iff ψ is analytic, and then the unbiased estimator is unique — hence automatically the UMVUE. The same uniqueness-by-transform argument runs through the Laplace transform for the exponential family.

The “uniqueness $\Rightarrow$ UMVUE” shortcut

Whenever a series- or transform-matching argument produces the *unique* unbiased estimator, it is the UMVUE with nothing further to prove: there is no competitor. Worth one sentence whenever it applies.

Companions: [PI_Bayes_Crash_Course.pdf](#), [PI_Testing_Crash_Course.pdf](#), [PI_Exam_Revision.pdf](#),
[PI_Class_Exercises_Solved.pdf](#), [PI_Assignment_Solutions.pdf](#), [PI_Mock_Exam.pdf](#).